

PRAGATHI DURGA RAJARAJAN

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📍 Lafayette, Indiana

RESEARCH INTERESTS

Interpretable and reliable machine learning, with a focus on continual and sample-efficient learning in multimodal and vision systems.

EDUCATION

Ph.D. in Computer Science, Purdue University August 2026 – Present

- **Coursework:** CS 571 Artificial Intelligence, CS 580 Algorithm Design, Analysis, And Implementation

B.S. in Computer Science, University of Texas at San Antonio August 2022 – May 2026

- *Honors Graduate with Distinction* | GPA: 3.98/4.0
- Concentrations: Data Science, Software Engineering | Minor: Mathematics

RESEARCH EXPERIENCE

Vision and AI Lab | Undergraduate Research Fellow Spring 2025 – Summer 2026

Advisor: Dr. Amanda Fernandez

- **Honors Thesis:** Investigated whether data pruning methods generalize from image classification to semantic segmentation, systematically evaluating their effects on data efficiency, training cost, and model performance across tasks. [T01, P01]
- Independently developed a SAM2-based multi-class materials segmentation repository, implementing adaptive point prompting, class-wise prediction aggregation, and experimentation infrastructure for continued lab research. [P03]
- Evaluated segmentation methods across materials-science datasets and architectures, including U-Net variants, DeepLabV3, PRNet, and SAM2; contributed model refinements, quantitative analyses, and multimodal defect-detection experiments resulting in a CVPR 2025 MM4Mat Workshop spotlight paper. [W01, P03]

Engaging Computing Group | Undergraduate Research Assistant Spring 2024 – Summer 2026

Advisor: Dr. Fred Martin

- Designed, developed, and empirically evaluated interactive AI education systems—including *FaunaForest*, *IntoTheRabbitHole*, *AI-See*, and *RuleRunners*—for teaching decision trees, search algorithms, image recognition, and rule-based AI to K–12 students. [C02, C04, C07, C08, P06, P07]
- Analyzed and annotated student interactions with LLM agents in *DoppelBot*, contributing CS education perspectives to study framing and analysis of how middle-school students identify AI agents in live collaborative settings; contributed to an EMNLP 2026 Main Conference paper. [C01]
- Conducted empirical studies with middle-school students, analyzing performance, interaction patterns, and feedback to investigate learning outcomes and design principles for interactive AI education systems. [C02, C04, C05, C07, C08]
- Led the design and instruction of four modules—*Image Recognition*, *Search Algorithms*, *Decision Trees*, and *Rule-based AI*—for UTSA’s *AI for Everyone* summer camp, integrating interactive tools and hands-on activities to support AI literacy. [C04, C05]

Systems and Networks Lab | Undergraduate Research Intern Summer 2023 – Spring 2026

Advisor: Dr. Rajendra Boppana

- Investigated out-of-distribution generalization in ML-based IoT malware detection, developing a reproducible ML experimentation pipeline for cross-dataset evaluation across models, dataset combinations, and feature configurations. [P04]
- Investigated ML-based ransomware detection by using benign resource-intensive workloads to safely simulate ransomware-like system behavior, collecting system telemetry and evaluating models for distinguishing simulated attacks from normal activity. [P08]

Advisor: Dr. Palden Lama

- Co-developed a distributed DNN inference pipeline using PyTorch RPC to deploy model shards across Raspberry Pi and GPU nodes, and identified Pareto-optimal partition points balancing inference latency and throughput. [C06, P05]
- Designed and developed a benchmarking repository for CNNs to measure resource utilization on edge devices. [P05]

TEACHING EXPERIENCE

Purdue University | Graduate Teaching Assistant, CS 180

Fall 2026 – Present

- Lead lab sections for *Problem Solving and Object-Oriented Programming*, introducing programming exercises and providing hands-on guidance; hold office hours and proctor course exams.

PUBLICATIONS

Peer-Reviewed Conference Papers

- C01. Dan Schumacher, **Pragathi Durga Rajarajan**, Haven Kotara, Roman Rendon, Kosi Atupulazi, Deepti Tagare, Ismaila Temitayo Sanusi, Fred G. Martin, and Anthony Rios. 2026. Detecting AI Impostors: How Do Middle Schoolers Identify LLM Agents in a Live Collaborative Setting? In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026). Association for Computational Linguistics. <https://arxiv.org/abs/2608.30948> [Accepted – To Appear]
- C02. **Pragathi Durga Rajarajan** and Fred Martin. 2026. RuleRunners: A Two-Player Collaborative Software Tool for Introducing Rule-Based AI Systems to Middle School Students. In Proceedings of the 2026 ACM Virtual Global Computing Education Conference (SIGCSE Virtual 2026). Association for Computing Machinery. [Accepted – To Appear]
- C03. Jonathan Perry, **Pragathi Durga Rajarajan**, and Amanda S. Fernandez. 2026. An Empirical Study of Sample Utility with Preferential Pruning. SPIE Defense + Commercial Sensing, Machine Learning on Challenging Data (MLCD 2026). <https://doi.org/10.1117/12.3098197>
- C04. **Pragathi Durga Rajarajan** and Fred Martin. 2026. On Teaching Image Recognition to Children at a Summer Camp. In Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1 (SIGCSE TS 2026), Vol. 1. Association for Computing Machinery, New York, NY, USA, 887–893. <https://doi.org/10.1145/3770762.3772635>
- C05. Kayleigh Stallings, Nicole Tian, Elif Yayla Ercek, Haven Kotara, Devin Marinelli, **Pragathi Durga Rajarajan**, Dan Schumacher, Ismaila Temitayo Sanusi, and Fred Martin. 2026. AI for Everyone: Engaging Middle Schoolers through Collaborative, Ethical, and Multimodal AI Learning. In Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1 (SIGCSE TS 2026), Vol. 1. Association for Computing Machinery, New York, NY, USA, 1012–1018. <https://doi.org/10.1145/3770762.3772627>
- C06. Adiba Masud, Nicholas Foley, **Pragathi Durga Rajarajan**, and Palden Lama. 2025. Where to Split? A Pareto-Front Analysis of DNN Partitioning for Edge Inference. IEEE 11th International Conference on Edge Computing and Scalable Cloud (EdgeCom), New York, United States. <https://doi.org/10.1109/EdgeCom66327.2025.00012>
- C07. **Pragathi Durga Rajarajan** and Fred Martin. 2025. IntoTheRabbitHole: A Web Application for Teaching Middle School Students About Search Algorithms. In Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1 (ITiCSE 2025). ACM, New York, NY, USA, 340–346. <https://doi.org/10.1145/3724363.3729080> [Best Paper Nominee]
- C08. **Pragathi Durga Rajarajan**, Adrian Cisneros and Fred Martin. 2025. FaunaForest: A Novel Software Tool for Teaching Decision Trees to Middle School Students. IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA, 2025, pp. 1-8, doi: [10.1109/ISEC64801.2025.11147419](https://doi.org/10.1109/ISEC64801.2025.11147419).

- C09. Cesar Hinojosa, Priyanka Kumar, **Pragathi Durga Rajarajan**, and Fred Martin. 2025. TrainYourSnakeAI: A Novel Tool to Teach Reinforcement Learning to Middle School Students. In Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE-TS 2025). ACM, New York, NY, USA, 506–512. <https://doi.org/10.1145/3641554.3701907>

Non-Archival Workshop Papers (Peer Reviewed)

- W01. Logan Robinson, Pedro Davila, Sergio Contreras, **Pragathi Durga Rajarajan**, Daniel Castillo, Michelle Voges, Cyana Zaragosa, Patrick Warren, Elizabeth S. Sooby, Amanda S. Fernandez. 2025. Evaluation and a New Benchmark for Defect Detection in Materials Fabrication. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Multimodal Models for Materials (MM4Mat), Nashville TN. [CVPR_MM4Mat_spotlight.pdf](#)

Theses

- T01. **Pragathi Durga Rajarajan**. 2026. *Data Pruning for Optimization of Artificial Intelligence Model Training and Generalizability Across Tasks*. Undergraduate Honors Thesis, University of Texas at San Antonio. <https://hdl.handle.net/20.500.12588/7774>

Selected Posters

* archival conference posters

- P01. **Pragathi Durga Rajarajan**, and Amanda Fernandez. 2026. Data Pruning for Optimization of AI Model Training and Insights Into Interpretability. UTSA Office of Undergraduate Research Spring 2026 Showcase.
- P02. * Priyanka Kumar, Panhapiseth Lim, **Pragathi Durga Rajarajan**, and Phillip Driscoll. 2026. Learning AI Ethics with EvolveMoralMaze: An Analysis of Student Outcomes and Misconceptions. In Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE-TS 2026), St. Louis, MO, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/37-70761.3777224>
- P03. **Pragathi Durga Rajarajan** and Amanda Fernandez. 2025. Multimodal Few-Shot Image Segmentation for Defect Detection. UTSA College of AI, Cyber, and Computing H-E-B Community Innovation Scholars and Team Research Projects Showcase.
- P04. * **Pragathi Durga Rajarajan**. 2025. Fast Attack Detection in Internet of Things (IoT) Devices Utilizing Machine Learning and Network Traffic Data. RSA Security Scholar Poster Exhibition. RSA Conference. <https://www.rsaconference.com/rsac-programs/security-scholar/poster-boards>
- P05. Nicholas Michael Foley, **Pragathi Durga Rajarajan**, Adiba Masud, and Palden Lama. 2025. Benchmarking Distributed DNN Inference in an Edge Environment. UTSA Office of Undergraduate Research Spring 2025 Showcase.
- P06. **Pragathi Durga Rajarajan**. 2025. FaunaForest: From Prototype to Paper. UTSA Honors College Experiential Learning Fair (Honors ELF Spring 2025). [1st Place in Research Track]
- P07. **Pragathi Durga Rajarajan** and Fred Martin. 2025. Designing Effective Software Tools for Enhancing K-12 AI Education. San Antonio Honors Research Symposium. [1st Place Winner]
- P08. **Pragathi Durga Rajarajan** and Anusha Abdulla. 2023. Using Simulated Data to Detect Ransomware. University of Texas at Austin Gateway to Graduate Studies in Science (G2S2 2023).

TEACHING AND MENTORSHIP

Purdue University | Graduate Teaching Assistant, CS 18000

Fall 2026 – Present

- Lead two weekly lab sections totaling approximately 50 students for *Problem Solving and Object-Oriented Programming*, reviewing course concepts, guiding programming exercises, and answering conceptual and technical questions.

- Provide individualized support during office hours by explaining programming concepts through examples and visualizations and helping students identify and reason through logical errors.

Research Mentorship | Research Mentor

Spring 2025 – Present

- Mentored high school students conducting research on large language models and guided mentees in developing research papers and posters presented at **IEEE ISEC 2025**.

AWARDS AND HONORS

Best Paper Nominee, ACM ITiCSE 2025

Summer 2025

- First-author paper selected as one of four Best Paper nominees (out of 99 accepted papers) at the ACM Innovation and Technology in Computer Science Education (ITiCSE), a flagship international venue for computing education research.

Microsoft Emerging Leaders (Asynchronous Cohort)

Summer 2025

- Selected for a competitive leadership development program focused on cultivating technical leadership, confidence, and personal branding for future innovators in technology.

UTSA Golden Feather Award

Spring 2025

- One of UTSA's two highest student honors, recognizing exceptional leadership, academic excellence, and transformative contributions that leave a lasting institutional legacy.

1st Place Poster, UTSA Honors College Experiential Learning Fair

Spring 2025

- Awarded first place in the Research track for presenting AI education research at the UTSA Honors College Experiential Learning Fair, a showcase highlighting experiential learning projects.

1st Place Poster, San Antonio Honors Research Symposium

Spring 2025

- Awarded first place for research on AI education at a citywide symposium featuring top honors students from six universities and colleges in San Antonio, Texas.

UTSA School of Data Science Undergraduate Research Fellow

Spring 2025

- Selected for the inaugural cohort of UTSA's School of Data Science fellowship program supporting high-impact undergraduate research in AI and data science.

RSAC Security Scholar

Fall 2024

- Sole UTSA student selected to represent the university at the RSA Conference 2025, joining a national cohort of distinguished cybersecurity scholars.

LEADERSHIP AND STUDENT ORGANIZATIONS

UTSA Honors College Research Student Panel | Panelist

Spring 2025

- Invited as one of six student panelists to share undergraduate research experiences and discuss strategies for pursuing research opportunities to Honors College students.

CodeQuantum Hackathon | Lead Design/Media Organizer

Summer 2024 – Spring 2026

- Led the Design/Media Team for CodeQuantum, developing promotional materials and a standardized outreach toolkit that supported 150 participants and 6 sponsors while strengthening the hackathon's visibility and long-term branding.

Rowdy Creators | Vice President, Media Director

Fall 2023 – Summer 2025

- Led media and outreach strategy for Rowdy Creators, directing officers and promotional initiatives that increased kickoff attendance from approximately 20 to 100 participants while strengthening member engagement and event visibility.

SKILLS

Programming: Python, Java, C, Bash, JavaScript, SQL, HTML/CSS

Machine Learning: PyTorch, TensorFlow, scikit-learn, computer vision, semantic segmentation, data pruning, distributed inference

Systems & Tools: Linux, Docker, PyTorch RPC, Zeek, Argus, Wireshark, Git/GitHub

Languages: English (fluent), Tamil (fluent), Spanish (intermediate)